Practice runs and test results

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use repeated trials to explain and support your design.

SUCCESS CRITERIA - I can...

1. Measure the actual course distance and run times.
2. Calculate average speed and describe variation.
3. Choose a final change supported by results from repeated trials.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “mean” to explain your answer. Sentence starters are optional.

trial: one run or repetition of a test

mean: sum of the values divided by the number of values

evidence: observations or measurements that support an idea

01 Warm-up: bring back an earlier idea

At unchanged wheel rpm, circumference becomes **5** times as large. **Compare ideal travel speed** and **state one real-world limitation**.

Word help: circumference = distance around the edge of a circle

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: mean time = sum of trial times/number of trials
average speed = distance/time.
Measure course length; keep release and timing consistent. Retain every trial and required build records.

Worked reasoning: Supplied **3 m** runs take **2.0**, **2.4**, **2.1 s**. Report all trials; their mean time is **2.17 s**. “Fastest run **2.0 s**” answers a different question from typical performance.

Your turn: Make a trial table and **mark both mean** and fastest time.

Sentence starter: I notice ... . This supports ... because ... .

Trial	Distance (m)	Time (s)	Speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Complete a final labeled sketch or attach a photo with labels. **Name a feature** that supports straight motion and one that supports reliable electrical operation.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

04 Apply the physics

A supplied practice run covers **4 m** in **5 s**. **Calculate average speed**. Then **use the table** for your own measured course length and three actual run times. Keep the sample calculation separate from your team’s data.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Trial	Distance (m)	Time (s)	Speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

One run stops early after a loose connection. Should you erase it? **Explain how** to **record it fairly** and **decide whether** it belongs in a speed comparison.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

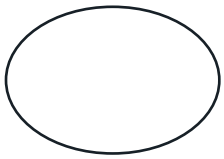
06 Review: connect an earlier idea

A magnet moves through a **60**-turn coil. Propose two controlled changes that could increase induced-emf magnitude and **explain each**.
Label the field or flux and **identify what changes**.

Word help: flux = a measure of magnetic field passing through a chosen area

Sentence starter: I think … . I can point to …, which shows …, so … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

07 Exit check

Name the final design change you will keep and cite one specific observation or measurement supporting it.

Sentence starter: My design includes … . It does what is needed by … .

Choice / claim	Actual evidence	Physics explanation
____	____	____
____	____	____

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Return to the worked example in problem **2**. **Compare this set** with **2.1**, **2.1**, **2.1 s**. Discuss speed and consistency rather than declaring one winner from the fastest run alone.

Sentence starter: Both … . They differ in … because … .

Trial	Distance (m)	Time (s)	Speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: ① ② ③